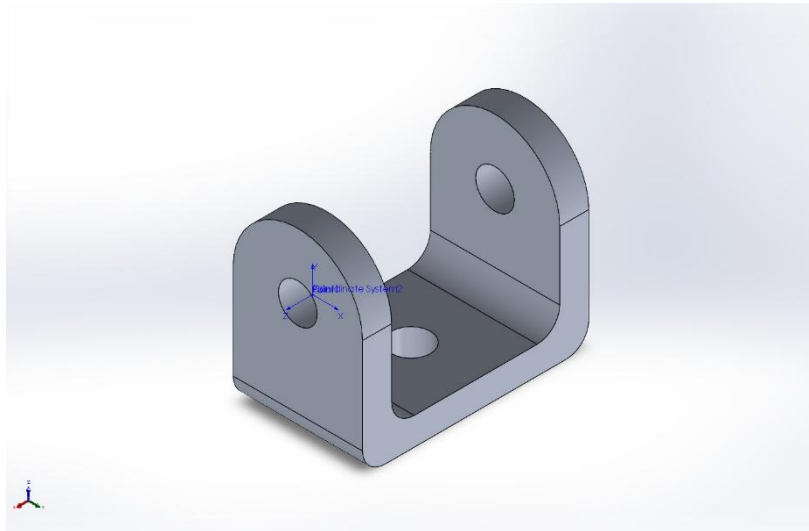




Description

This study analyzes a U-shaped aluminum 7075-T6 bracket. Determine the natural frequencies of the bracket to ensure dynamic stability for UAV loading conditions.

Simulation of U-Bracket for UAV Wing Attachment

Date: Sunday, November 16, 2025

Designer: Kevin Thomas Fernandez

Study name: Bracket Frequency study

Analysis type: Frequency

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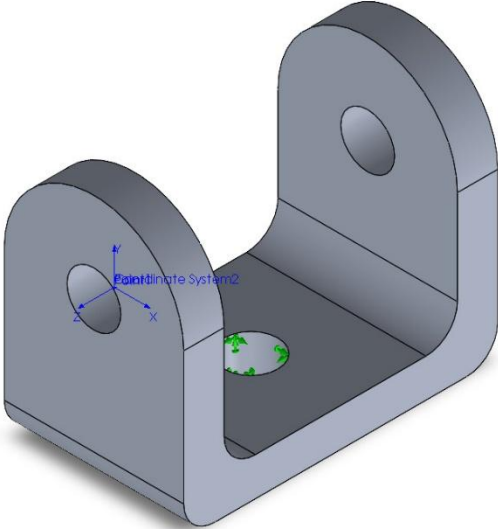
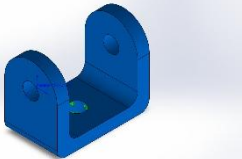


Assumptions

The frequency analysis assumes the bracket is a single linear elastic, isotropic 7075-T6 aluminum body without damping or material nonlinearity. The geometry is identical to that used for the static analysis, and no additional parts, fasteners, or surrounding structure are included in the model. The bracket's base mounting face is fully fixed to represent rigid attachment, while no external forces or moments are applied since the analysis seeks to determine natural modes of vibration. The modal study uses undamped eigenvalue extraction with the FFEPlus solver to compute the first five natural frequencies. Effects such as aerodynamic loading, operational vibration input, thermal gradients, contact interactions, and added mass from bolts or assemblies are not considered. A high-quality curvature-based solid mesh is used to ensure resolution of dynamic behavior. This simplified setup is intended to isolate the bracket's inherent modal characteristics for comparison against the minimum frequency requirement.



Model Information

 Model name: U-Bracket for UAV Wing Attachment Current Configuration: Default			
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Cut-Extrude4 	Solid Body	Mass:0.146886 kg Volume:5.22727e-05 m^3 Density:2,810 kg/m^3 Weight:1.43948 N	U-Bracket for UAV Wing Attachment.SLDPRT Nov 16 18:13:01 2025

Study Properties

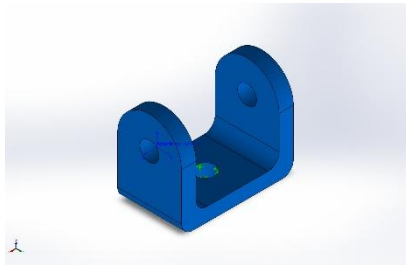
Study name	Bracket Frequency study
Analysis type	Frequency
Mesh type	Solid Mesh
Number of frequencies	5
Decouple the mixed free body modes	Off
Solver type	FFEPlus
Soft Spring:	Off
Incompatible bonding options	Automatic
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off

Units

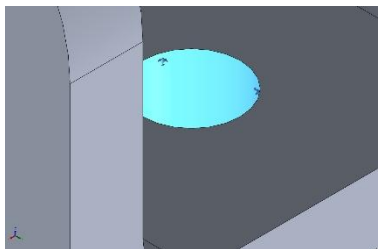
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²



Material Properties

Model Reference	Properties	Components
	Name: 7075-T6 (SN) Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 5.05e+08 N/m ² Tensile strength: 5.7e+08 N/m ² Mass density: 2,810 kg/m ³ Elastic modulus: 7.2e+10 N/m ² Poisson's ratio: 0.33 Thermal expansion coefficient: 2.36e-05 /Kelvin	U-Bracket (U-Bracket for UAV Wing Attachment)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry



Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	3.74025 mm
Minimum element size	3.74025 mm
Mesh Quality	High

Mesh information - Details

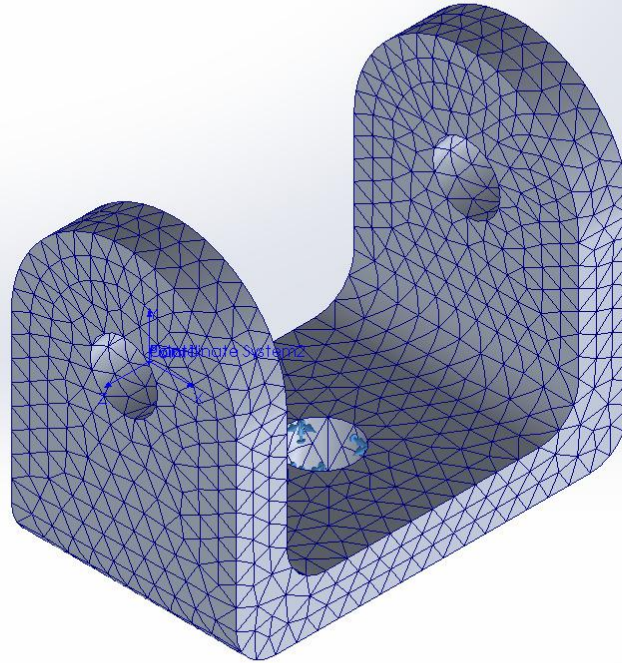
Total Nodes	15260
Total Elements	9353
Maximum Aspect Ratio	3.9365
% of elements with Aspect Ratio < 3	99.9
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:01
Computer name:	STARS

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Model name: Part5
Study name: Bracket Frequency study(-Default-)
Plot type: Mesh Quality1

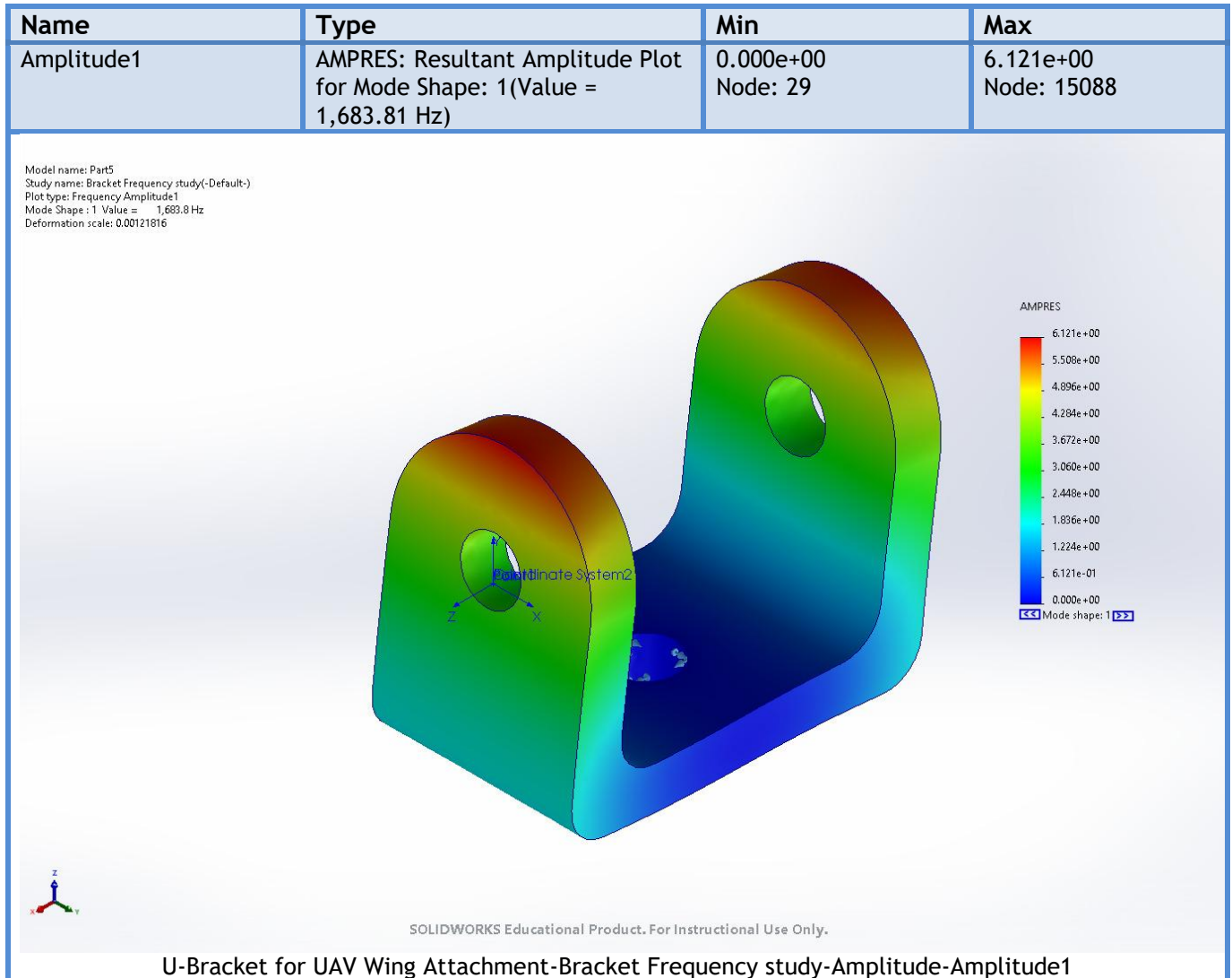


SOLIDWORKS Educational Product. For Instructional Use Only.

U-Bracket for UAV Wing Attachment-Bracket Frequency study-Quality-Quality1

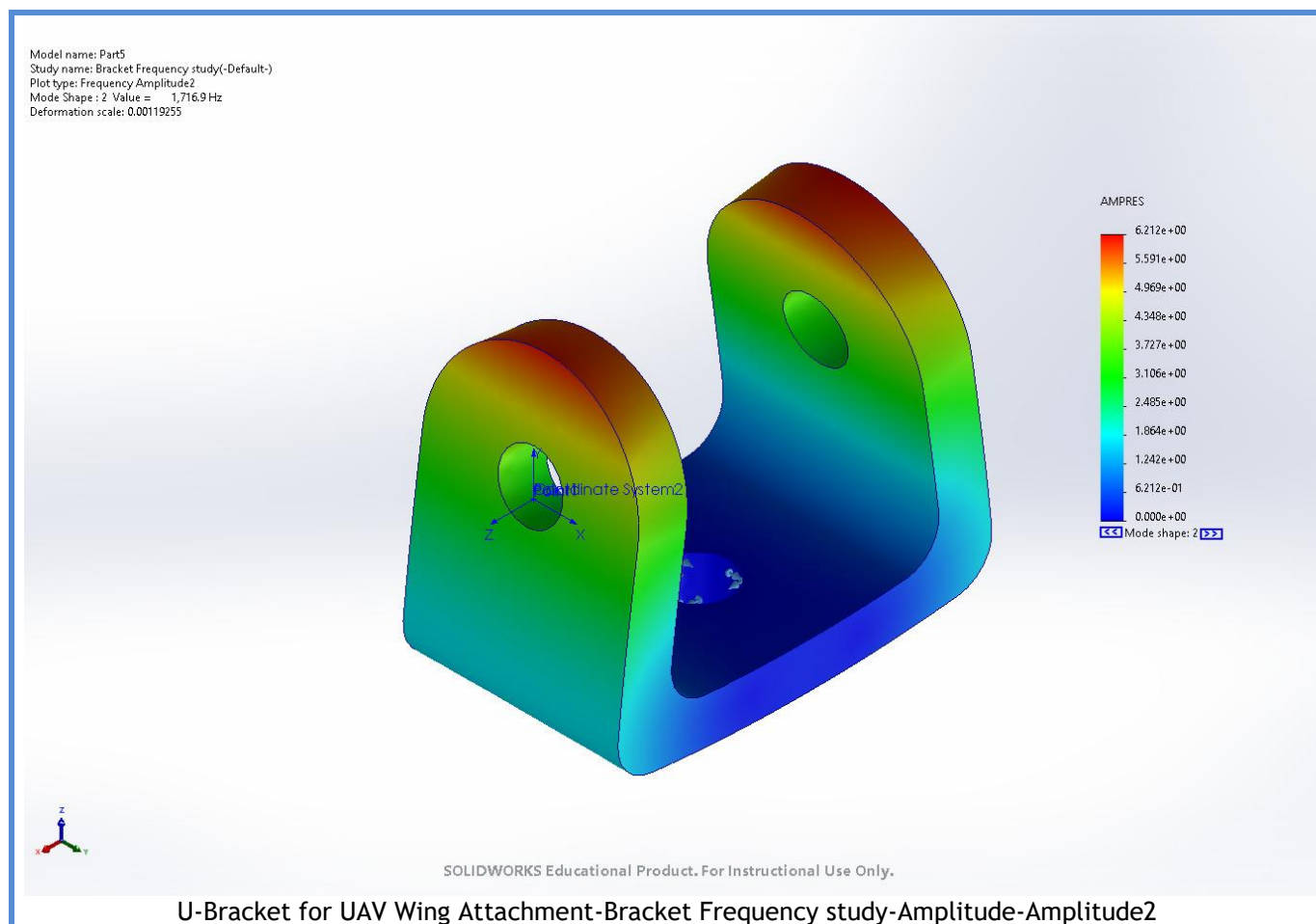


Study Results

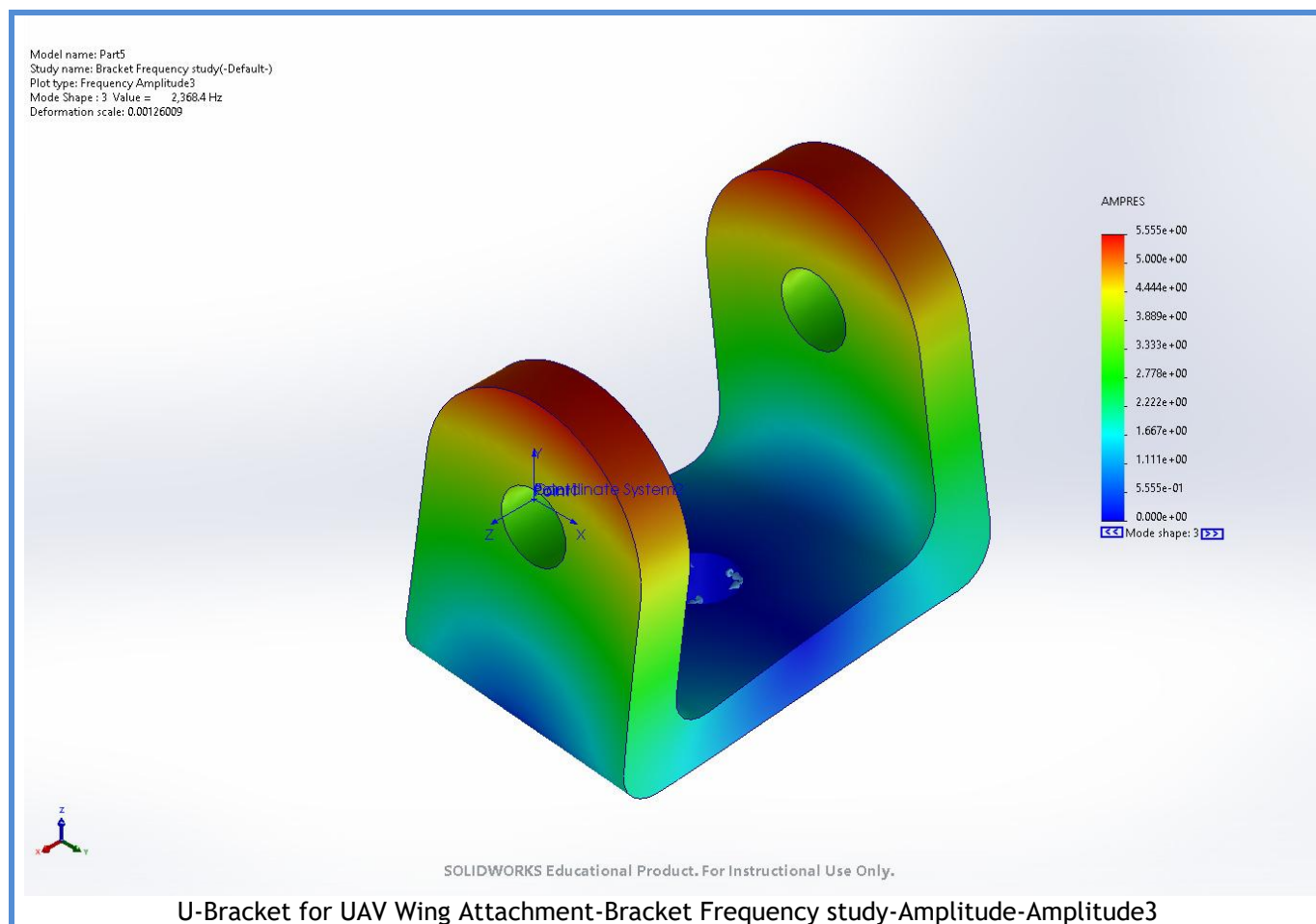


Name	Type	Min	Max
Amplitude2	AMPRES: Resultant Amplitude Plot for Mode Shape: 2(Value = 1,716.87 Hz)	0.000e+00 Node: 29	6.212e+00 Node: 15052

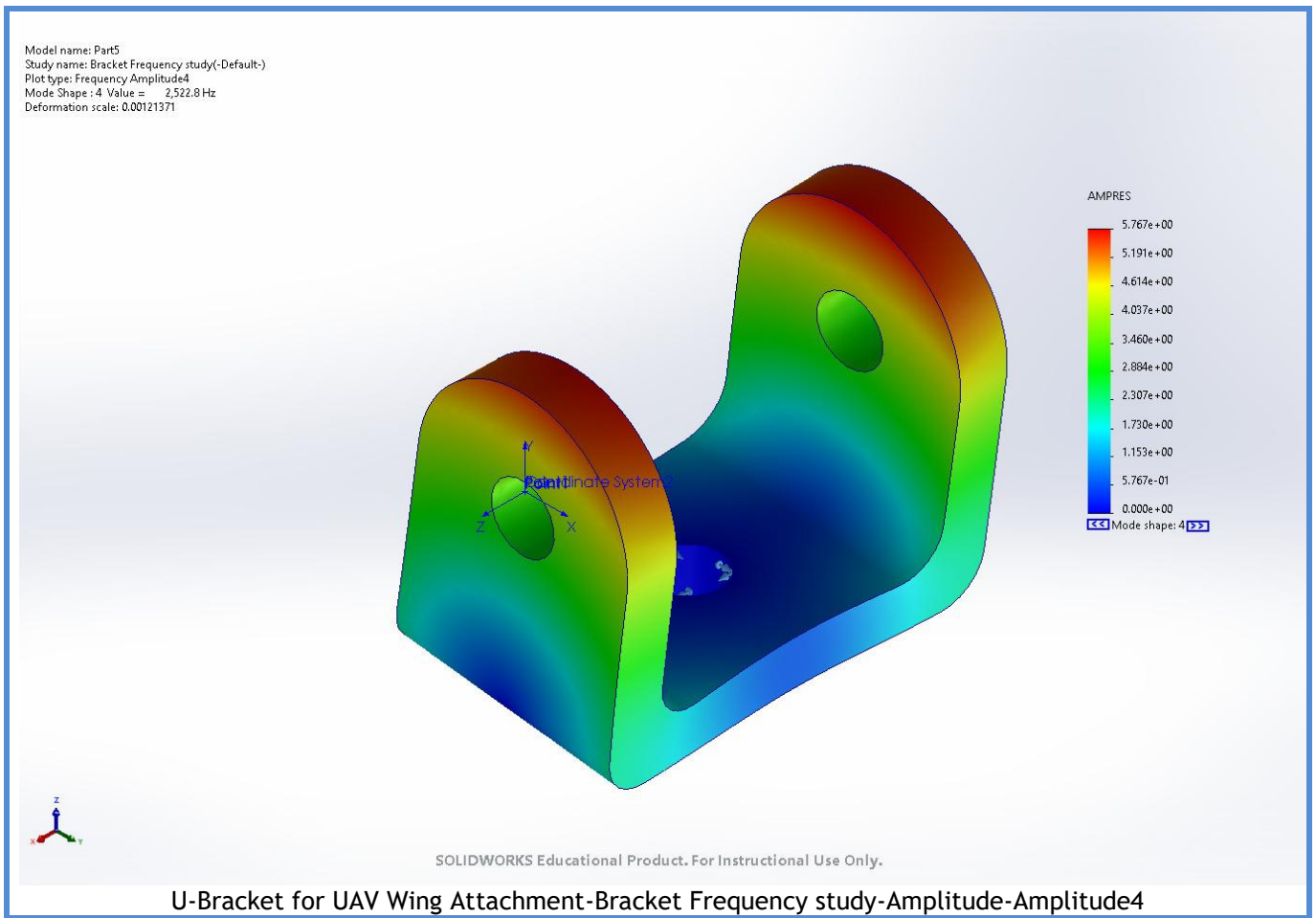




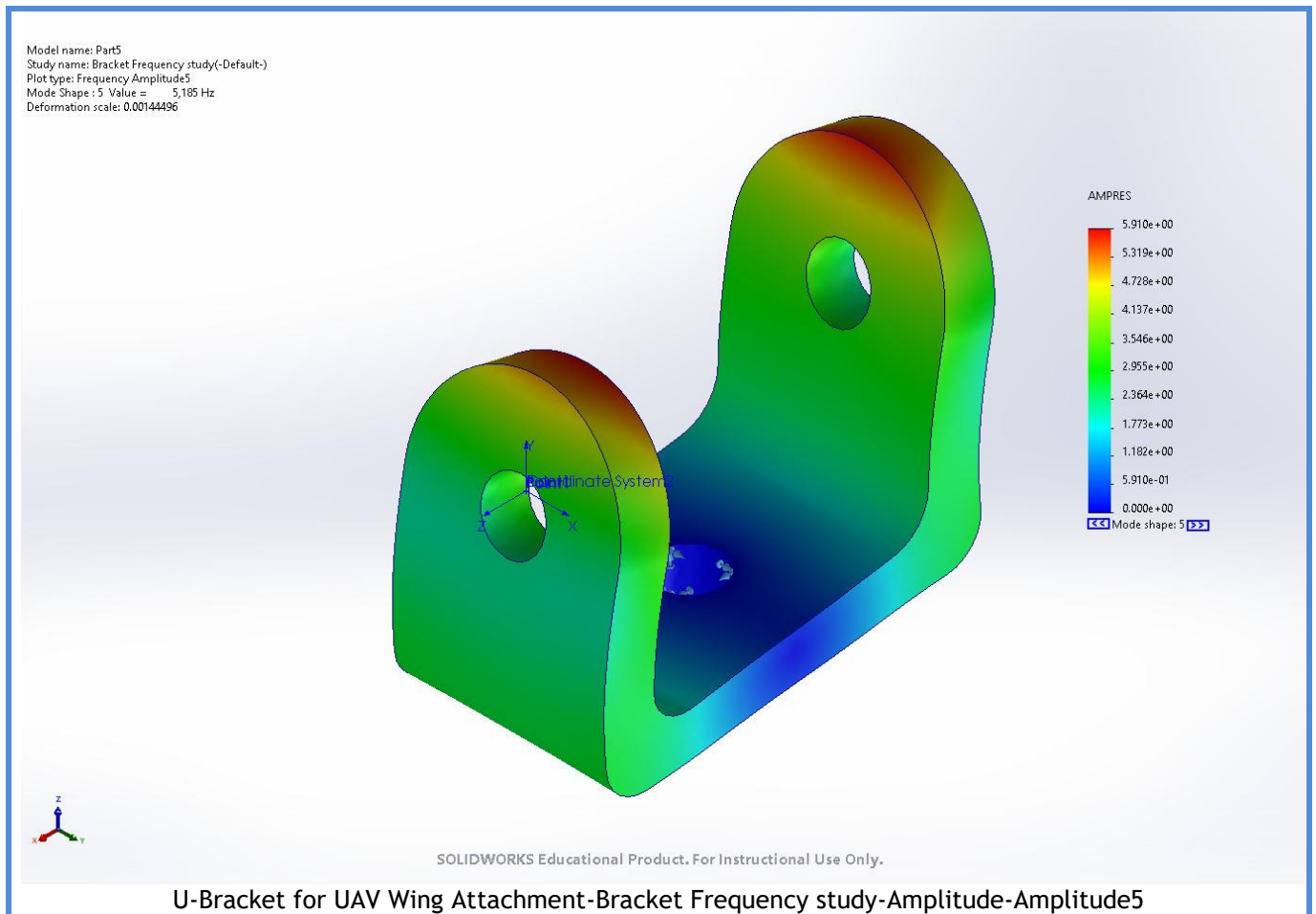
Name	Type	Min	Max
Amplitude3	AMPRES: Resultant Amplitude Plot for Mode Shape: 3(Value = 2,368.44 Hz)	0.000e+00 Node: 29	5.555e+00 Node: 14721



Name	Type	Min	Max
Amplitude4	AMPRES: Resultant Amplitude Plot for Mode Shape: 4(Value = 2,522.8 Hz)	0.000e+00 Node: 29	5.767e+00 Node: 15108



Name	Type	Min	Max
Amplitude5	AMPRES: Resultant Amplitude Plot for Mode Shape: 5(Value = 5,184.98 Hz)	0.000e+00 Node: 29	5.910e+00 Node: 14721



Mode List

Frequency Number	Rad/sec	Hertz	Seconds
1	10,580	1,683.8	0.00059389
2	10,787	1,716.9	0.00058246
3	14,881	2,368.4	0.00042222
4	15,851	2,522.8	0.00039638
5	32,578	5,185	0.00019286

Mass Participation (Normalized)

Mode Number	Frequency(Hertz)	X direction	Y direction	Z direction
1	1,683.8	0.32356	1.9816e-08	1.8615e-05
2	1,716.9	3.3217e-05	3.423e-09	0.16882
3	2,368.4	3.47e-07	3.3405e-06	8.2268e-13
4	2,522.8	2.3487e-08	0.39527	1.3714e-07



5	5,185	0.053967	6.0751e-09	2.5826e-05
		Sum X = 0.37756	Sum Y = 0.39527	Sum Z = 0.16886

Conclusion

The modal analysis shows that the bracket's first natural frequency is approximately 1684 Hz, which exceeds the minimum required 50 Hz by more than 33 times. This significant margin indicates that the bracket is exceptionally stiff and dynamically stable relative to expected UAV operational vibration environments. With all higher modes also occurring well above the 50 Hz threshold, the structure is not susceptible to resonance within any realistic flight conditions or propulsion-induced vibration spectra. Overall, the bracket meets and substantially surpasses all dynamic performance requirements and is fully suitable for integration into the UAV airframe.

